

# Auto Ticket Classification using Flow Designer

## Project Design Phase-II: Data Flow Diagram & User Stories

|                      |                                                |                       |                 |
|----------------------|------------------------------------------------|-----------------------|-----------------|
| <b>Date:</b>         | 03 October 2026                                | <b>Team ID:</b>       | SWTID-2026-5758 |
| <b>Project Name:</b> | Auto Ticket Classification using Flow Designer | <b>Maximum Marks:</b> | 4 Marks         |

### Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) visually represents the information flow within the system. It illustrates how a requester submits an IT incident request on the custom table (Incident Workflow), how Flow Designer evaluates keywords in the Short Description to assign Category and Subcategory, and how automated status notifications are routed to the user.

#### Level 0 - Context Diagram

| External Entity                             | Data Flow (Input)                          | System                                                | Data Flow (Output)                         | External Entity                       |
|---------------------------------------------|--------------------------------------------|-------------------------------------------------------|--------------------------------------------|---------------------------------------|
| <b>Requester / User (Student / Teacher)</b> | Ticket details (Caller, Short Description) | <b>Auto Ticket Classification using Flow Designer</b> | Classification result & Email notification | <b>Support Agent / Assigned Group</b> |

#### Level 1 - Detailed DFD

| Process                              | Input                                                        | Processing / Data Store                                   | Output                                        |
|--------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------|
| <b>1. Create Ticket</b>              | Short Description, Caller details                            | Custom Table: Incident Workflow [u_incident_workflow]     | Ticket record created (Auto Number generated) |
| <b>2. Trigger Flow</b>               | New record created & Category is empty                       | Flow Designer Trigger ('Auto Classify School IT Tickets') | Flow execution initiated                      |
| <b>3. Classify Ticket</b>            | Short Description keywords (WiFi, Projector, Password, Slow) | Flow Logic (If / Else If Keyword Conditions)              | Category & Subcategory populated              |
| <b>4. Route &amp; Email</b>          | Classified ticket record                                     | Flow Designer Update Record & Send Email Action           | Updated ticket record & Email sent to Caller  |
| <b>5. Agent Review &amp; Resolve</b> | Classified ticket                                            | IT Support Agent review & resolution                      | Resolved / Closed ticket state                |

### User Stories

| User Type   | Epic / Requirement  | USN          | User Story / Task                                                                                   | Acceptance Criteria                                                                                                   | Pri. | Rel. |
|-------------|---------------------|--------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------|------|
| Requester   | Ticket Creation     | <b>USN-1</b> | As a user, I can create a support ticket by entering the caller name and short description.         | Ticket created with unique auto-number; Category/Subcategory left blank for auto-fill.                                | High | Sp-1 |
| Requester   | Ticket Creation     | <b>USN-2</b> | As a user, I can view the status and classification of my submitted ticket.                         | User can view ticket number, assigned category, and current state.                                                    | High | Sp-1 |
| Flow/System | Automatic Trigger   | <b>USN-3</b> | As the system, I automatically trigger Flow Designer when a new ticket is created without category. | Flow 'Auto Classify School IT Tickets' triggers when Category is empty on Incident Workflow.                          | High | Sp-1 |
| Flow/System | Auto Classification | <b>USN-4</b> | As the system, I classify tickets into Network, Hardware, Access, or Performance based on keywords. | WiFi->Network/WiFi, Projector->Hardware/Projector, Password->Access/Forgot Password, Slow->Performance/Slow Computer. | High | Sp-1 |
| Flow/System | Notification        | <b>USN-5</b> | As the system, I send an email confirmation to the caller once ticket is created and classified.    | Automated email sent to Caller email address containing ticket details.                                               | High | Sp-2 |

|               |                          |              |                                                                                                     |                                                                                       |        |      |
|---------------|--------------------------|--------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|------|
| Flow/System   | Assignment & Routing     | <b>USN-6</b> | As the system, I can assign tickets to the relevant Assigned Group and Assigned To user.            | Assigned Group and Assigned To fields reference sys_user_group and sys_user properly. | High   | Sp-2 |
| Support Agent | Ticket Review            | <b>USN-7</b> | As a support agent, I can review auto-classified tickets and modify Category/Subcategory if needed. | Dependent choice fields enforce subcategory options based on selected category.       | High   | Sp-2 |
| Support Agent | Ticket Resolution        | <b>USN-8</b> | As a support agent, I can update ticket state from New/In Progress to Resolved or Closed.           | State updates properly and retains full audit history in update set.                  | High   | Sp-3 |
| Administrator | Deployment & Maintenance | <b>USN-9</b> | As an admin, I can capture all backend changes in an Update Set and export XML for migration.       | Update set 'Project Update Set' completed and exported to XML successfully.           | Medium | Sp-3 |

## Project Summary

The Auto Ticket Classification using Flow Designer project delivers an end-to-end ServiceNow no-code solution for school IT helpdesks. By capturing caller input on the custom 'Incident Workflow' table and utilizing Flow Designer keyword logic, tickets are automatically classified into exact Categories and Subcategories (Network, Hardware, Access, Performance). Automated email notifications are dispatched immediately, and all configurations are packaged into a completed Local Update Set for seamless deployment.